Curs 8

R.A.N.

BE Pol directă

CB Pol inversă

$I_c = \beta I_B$

Componente

↓

Polarizare

↓

P.S.F $I_c, U_{CE} > U_{BE}$ (NPN)

Stabilizator de tensiune serie



$U_o = ct$

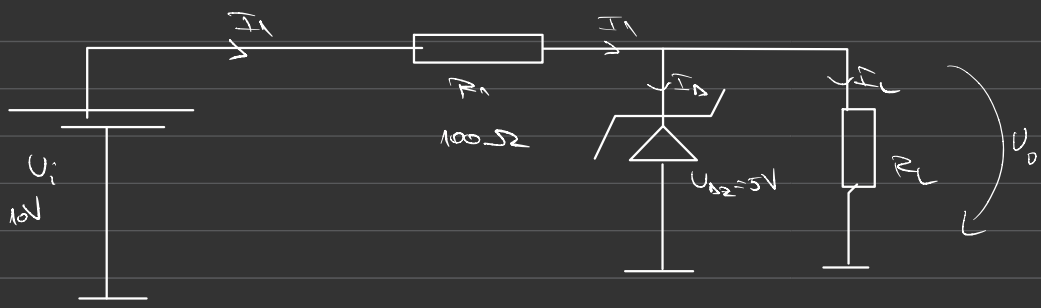
$U_o \neq f(U_i)$

$U_o \neq f(R_L)$

Pentru rețea: 100–270 V $= U_i$ (~AC)

19 V / 3 A $= U_o$ (~AC)

Stabilizator paralel

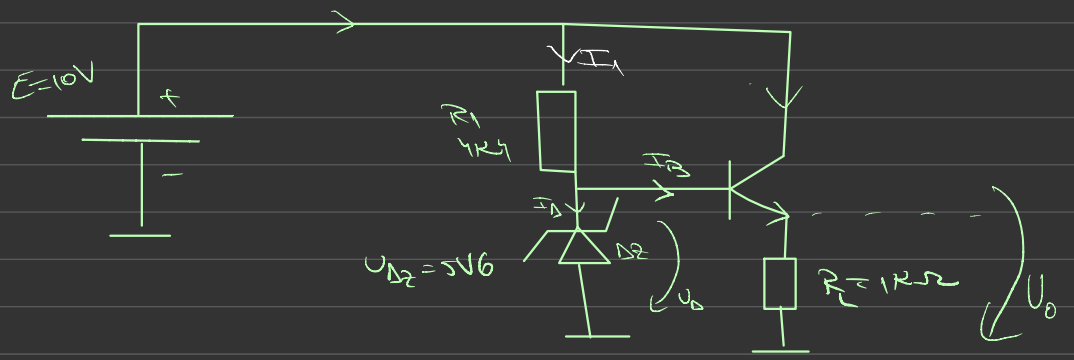


$I_1 = I_D + I_L$

$I_D > 0 \Rightarrow U_0 = U_{DZ} = ct$

$I_1 = \frac{U_i - U_{DZ}}{R_1} = 50\,mA$

R_L	I_L	I_D	I_1
1K	50 mA	45 mA	50 mA
0,1K	10 mA	40 mA	50 mA
∞	0	50 mA	50 mA
10Ω	500 mA		



pp ca T RAN, DZ cond inv ($I_D > 0$) $\Rightarrow U_D = U_{DZ} = 5,6\,V$

$I_C = \beta I_B$ $\quad U_{BE} \simeq 0,6\,V$

$U_D = U_{BE} + U_0 \quad \Rightarrow U_0 = U_D - U_{BE} = 5,6 - 0,6 = 5V$

$I_1 = I_D + I_B$

$U_0 = R_L I_E \simeq R_L I_C \quad \Rightarrow \quad I_C = 5mA \Rightarrow I_B = \frac{I_C}{\beta} = 0,05\,mA$

$E = R_1 I_1 + U_D \quad \Rightarrow I_1 = \frac{4,4V}{4K4} = 1mA$

$I_D = 1mA - 0,05\,mA = 0,95\,mA > 0$

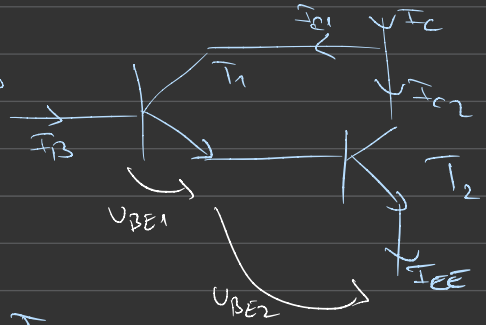
R_L	I_L	I_D	I_1	$I_i = I_1 + I_D \simeq I_1 + I_L$
1 kΩ	5 mA	0,95 mA	1 mA	6 mA
0,5 kΩ	10 mA	0,9 mA	1 mA	11 mA
∞	0	1 mA	1 mA	1 mA

$I_{L\,max} = 100$ mA

Pour $I_{L\,max}$ on veut $I_{i\,max} \Rightarrow R_1$ maximale

Transistors composés

T. DARLINGTON

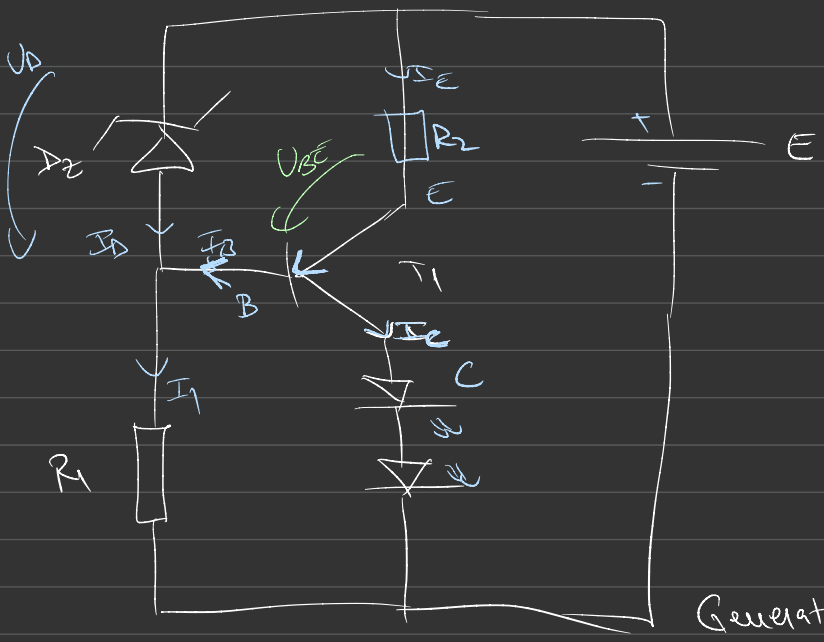


$I_C = I_{C1} + I_{C2}$

$I_{B_2} = I_{E1} = I_{B_1}(\beta+1) = I_B(\beta+1)$

$I_C = \beta I_{B_1} + \beta I_{B_2} = \beta I_B + \beta(\beta+1) I_B \simeq \beta^2 I_B$

$U_{BE} = U_{BE1} + U_{BE2}$



Generator curent constant

D_Z – cond inversă $(I_D > 0)$

$U_D = U_{DZ}$

$U_D = I_E R_2 + U_{EB} \Rightarrow I_E = \dfrac{U_D - U_{EB}}{R_2}$ (ct) (U_{DZ} = ct, $U_{EB} \approx 0{,}6$)

$I_C \approx I_E = \dfrac{U_{DZ} - U_{EB}}{R_2}$

$I_D = I_1 - I_B > 0 \Rightarrow \dfrac{E - U_{DZ}}{R_1} - I_B > 0$

$I_B = \dfrac{I_C}{\beta}$

TRAN : $U_{EC} > 0{,}6\ V$

$$U_{EC} = E - I_C R_2 - U_{AL} \geq 0{,}6\,V$$

Ex: $E = 12\,V$ $\quad I_C = 10\,mA$ $\quad U_{AZ} = 3{,}3$

$$\Rightarrow R_2 = \frac{U_{AZ} - U_{EB}}{I_C} = 270\,\Omega$$

$$I_B = \frac{I_C}{\beta} = 91\,mA$$

Adoção $I_1 = 1\,mA$

$$R_1 = \frac{E - U_{AZ}}{I_1} = 8\,K\Omega$$

$$U_{AZ\,max} = E - R_2 I_C - U_{EC\,min} = 12 - 2{,}7 - 0{,}6\,V =$$

$$= 8{,}7\,V$$